

Complete HTML Notes

Clean, structured reference notes — HTML fundamentals to forms, tables, iframes, and meta tags

Quick Roadmap

HTML basics → document structure → tags & elements → block/inline → semantic HTML → formatting → links & images → forms & validation → lists → tables → iframes → useful elements → attributes → meta tags

1. Introduction to HTML

What is HTML?

HTML stands for HyperText Markup Language. It is used to create web pages. HTML tells the browser what to display, where to display it, and how elements are organized. HTML is a markup language, not a programming language.

What is HyperText?

HyperText means text that contains links to another page. Example: `<a href="page.html">Click Here</a>`.

What is Markup Language?

Markup means using tags to structure content. Example: `<h1>Hello</h1>`, where `<h1>` is a markup tag.

Why Learn HTML?

HTML is used in websites, web applications, emails, landing pages, and dashboards. Without HTML, a webpage cannot be structured.

House Analogy

Blueprint → HTML
Painting → CSS
Electricity → JavaScript

2. HTML Versions

Versions

Version	Year
HTML 1.0	1993
HTML 2.0	1995
HTML 3.2	1997
HTML 4.01	1999
XHTML	2000
HTML5	2014

Current Version

HTML5 is the modern HTML standard commonly taught as HTML5.

3. Advantages of HTML

Advantages

- Easy to learn
- Platform independent
- Free
- Supported by all major browsers
- SEO friendly

4. Browser and How It Works

What is a Browser?

A browser reads HTML files. Examples: Chrome, Edge, Firefox, Safari.

How Browser Works

HTML File → Browser reads HTML → Browser parses/converts HTML → Webpage is displayed

5. Basic HTML Structure

Basic Structure

```
<!DOCTYPE html>
<html>
<head>
</head>
<body>
</body>
</html>
```

<!DOCTYPE html>

Tells the browser that the document uses HTML5. Without a DOCTYPE, browsers may enter Quirks Mode.

<html>

The root element. Everything in the document is contained inside it.

<head>

Contains the title, CSS links, JavaScript links, meta tags, and icons. Its contents are generally not displayed as page content.

<body>

Contains the visible webpage content.

Complete Example

```
<!DOCTYPE html>
<html>
<head>
  <title>My First Page</title>
</head>
<body>
  Welcome to HTML
</body>
</html>
```

6. HTML Tree Structure

Root Element

<html> is the top-most/root element.

Parent and Child

```
<body>
  <h1>Hello</h1>
</body>
Parent: body
Child: h1
```

Nested Elements

```
html
  head
    title
```

```
■■■ body
  ■■■ h1
```

7. Container and Empty Elements

Container Elements

Have opening and closing tags. Examples: <html>, <head>, <body>, <div>, <p>, <h1>, <section>, <article>, <nav>, <header>, <footer>.

Empty Elements

Do not have closing tags. Examples:
, <hr>, .

Live Coding Example

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Page</title>
</head>
<body>
  <h1>Welcome Students</h1>
  <p>This is our first HTML class.</p>
</body>
</html>
```

8. Tags and HTML Elements

What is a Tag?

An HTML tag is a keyword enclosed in angle brackets (< >) that tells the browser how to structure or represent content.

Tag Syntax

```
<tagname>Content</tagname>
```

What is an HTML Element?

An HTML element is the complete structure consisting of an opening tag, content, and closing tag.

Formula

Element = Opening Tag + Content + Closing Tag

Example

```
<h1>Welcome to HTML</h1>
Opening tag: <h1>
Content: Welcome to HTML
Closing tag: </h1>
```

9. Block-Level and Inline Elements

Block-Level Elements

A block-level element normally starts on a new line and takes the available width. Examples include <h1> and <p>.

Inline Elements

An inline element normally does not start on a new line and takes only the space it needs. Example: HTML.

10. Common HTML Tags

<p>

Paragraph element. Example: <p>HTML is easy to learn.</p>

<h1> to <h6>

Heading levels from the highest/most prominent heading level h1 to h6.

<div>

Block-level generic container used to group elements.

Inline generic container used to group or style a small part of text.

<a>

Anchor element used to create hyperlinks.

Image element used to display an image. It is an empty element.

11. Semantic HTML

What is Semantic HTML?

Semantic HTML uses meaningful elements that describe the purpose of content, such as <header>, instead of using only generic <div> elements.

Advantages

- Better code readability
- Better SEO
- Better accessibility
- Easier maintenance

<header>

Represents introductory/top content, often containing a logo, site name, or navigation.

<footer>

Represents bottom/footer content such as copyright, contact details, or policies.

<nav>

Represents a navigation section.

<section>

Represents a thematic section of related content.

<article>

Represents independent, self-contained content such as a blog post or news article.

<aside>

Represents related content such as a sidebar, related links, categories, or advertisements.

<main>

Represents the primary content of the page. Generally one main element is used per page.

div vs Semantic Tags

div	Semantic tags
Generic container	Meaningful container
No specific meaning	Describe content purpose
Grouping/layout	Page structure
Example: <div>	Examples: <header>, <footer>, <section>

12. HTML Formatting Tags

What are Formatting Tags?

Formatting tags can change the presentation of text and/or communicate meaning.

Displays text in bold without indicating importance.

Displays text in bold by default and indicates strong importance.

b vs strong

	
Visual bold	Bold + semantic importance

<i>

Displays text in italic style.

Emphasizes text; browsers usually display it in italic.

i vs em

<i>	
Italic style	Italic + semantic emphasis

<u>

Underlines text.

<mark>

Highlights text.

<small>

Displays text smaller than surrounding text.

<sup>

Displays text as superscript. Example: x².

<sub>

Displays text as subscript. Example: H₂O.

Indicates deleted text, normally shown with a strike-through.

<ins>

Indicates inserted text, normally underlined.

13. Interactive HTML Tags

What are Interactive Tags?

Elements that allow users to interact with a webpage by clicking links, entering data, or submitting information.

<a>

Creates hyperlinks. Important attribute: href, which specifies the destination.

<button>

Creates a clickable button.

<input>

Receives user input. It is an empty element.

Common Input Types

Type	Purpose
text	Single-line text
password	Password input
email	Email address
number	Numeric input
date	Date picker
radio	Single selection
checkbox	Multiple selections
file	File upload
submit	Submit form
reset	Reset form

<textarea>

Creates a multi-line text input field. Example: <textarea rows="5" cols="30"></textarea>.

14. HTML Attributes

What are Attributes?

Attributes provide additional information about an HTML element and can modify behavior or functionality. They are written in the opening tag.

Syntax

<tagname attribute="value">Content</tagname>

Global Attributes

Common global attributes include id, class, title, style, lang, and hidden.

id

Assigns an identifier to an element. IDs are intended to be unique within a document.

class

Groups elements so the same class can be applied to multiple elements.

id vs class

id	class
Unique identifier	Can be reused
Usually targets one element	Can target multiple elements

title

Provides additional information, commonly shown as a tooltip.

style

Applies inline CSS directly to an element.

lang

Specifies the language of the document or element. Example: <html lang="en">.

hidden

Hides an element from normal rendering. Example: <p hidden>Hidden text</p>.

15. HTML Links

<a> Syntax

```
<a href="URL">Link Text</a>
```

href

Specifies the destination URL.

target

Value	Description
_self	Same browsing context (default)
_blank	New browsing context/tab
_parent	Parent browsing context
_top	Top-level browsing context

Examples

```
<a href="https://www.google.com">Visit Google</a>
<a href="https://www.google.com" target="_blank">Google</a>
<a href="about.html">About Us</a>
<a href="mailto:abc@gmail.com">Send Email</a>
<a href="tel:9876543210">Call Us</a>
```

16. HTML Images

Displays an image and is an empty element.

Syntax

```

```

src

Specifies the image file path or URL.

alt

Provides alternative text when an image cannot be displayed and supports accessibility.

width / height

Specify image dimensions.

Complete Example

```

```

Relative Path

```
 – image is referenced from the project structure.
```

Absolute Path

```
 – image is referenced by a full URL.
```

17. HTML Forms

What is an HTML Form?

A form collects information from users and can send it to a server for processing.

Common Uses

Registration, login, contact forms, feedback, online applications, and surveys.

<form>

Creates a form containing inputs, buttons, and other form controls.

Basic Example

```
<form>
  <label>Name:</label>
  <input type="text">
</form>
```

18. Form Validation

What is Form Validation?

The process of checking whether user-entered information is valid and complete before submission.

Why Validation?

- Prevent empty fields
- Check email format
- Accept suitable data types
- Improve data quality

- Improve user experience

required

Makes a field mandatory.

Email Validation

```
<input type="email" required>
```

Number Validation

```
<input type="number" required>
```

URL Validation

```
<input type="url">
```

pattern

Checks whether input matches a specified pattern. Example: `<input type="text" pattern="[A-Za-z]{3,20}">`

placeholder

Displays a hint inside an input field.

Form Example

```
<form>
  <label>Name</label>
  <input type="text" placeholder="Enter your Name" required>
  <br><br>
  <label>Email</label>
  <input type="email" placeholder="Enter your Email" required>
  <br><br>
  <label>Mobile</label>
  <input type="number" required>
  <br><br>
  <button>Submit</button>
</form>
```

19. HTML Lists

What is a List?

A list displays related items in an organized manner.

Types

Unordered list, ordered list, and description list.

Creates a bulleted list.

Represents an individual list item.

Creates a numbered list.

<dl>

Creates a description list.

<dt>

Defines a term in a description list.

<dd>

Provides the description of the term.

Example

```
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language</dd>
  <dt>CSS</dt>
  <dd>Cascading Style Sheets</dd>
  <dt>JS</dt>
  <dd>JavaScript</dd>
</dl>
```

20. HTML Tables

What is a Table?

A table displays information in rows and columns, such as student marks, employee details, product lists, and timetables.

<table>

Creates a table.

<tr>

Creates a table row.

<th>

Creates a header cell; browsers usually display it bold and centered by default.

<td>

Creates a data cell.

Basic Example

```
<table border="1">
  <tr>
    <th>Name</th>
    <th>Age</th>
  </tr>
  <tr>
    <td>John</td>
    <td>22</td>
  </tr>
  <tr>
    <td>David</td>
    <td>21</td>
  </tr>
</table>
```

th vs td

<th>	<td>
Header cell	Data cell
Bold by default	Normal text

21. Iframes

What is an iframe?

An iframe embeds another webpage, document, map, or video inside the current webpage.

Syntax

```
<iframe src="URL" width="500" height="300" title="Embedded content"></iframe>
```

Common Attributes

Attribute	Purpose
src	URL of embedded content
width	Width
height	Height
title	Description for accessibility

YouTube Example

```
<iframe width="500" height="300" src="https://www.youtube.com/embed/VIDEO_ID" title="YouTube video"></iframe>
```

Google Maps

Use the Google Maps Embed URL in the iframe src attribute.

iframe vs Hyperlink

iframe	Hyperlink
Embeds content inside current page	Navigates to another resource
Uses <iframe>	Uses <a>

22. Useful HTML Elements

HTML Comments

Comments are notes in source code and are not displayed as webpage content. Syntax: <!-- This is a comment -->

<details>

Creates a collapsible section.

<summary>

Provides the visible heading for a <details> element.

<progress>

Displays task progress. Example: <progress value="80" max="100"></progress>.

<meter>

Represents a scalar value within a known range, such as a battery level.

progress vs meter

progress	meter
Task progress	Measurement/value

progress	meter
Example: download progress	Example: battery level

<figure>

Groups an image, chart, or diagram with related content/caption.

<figcaption>

Provides a caption for a <figure>.

<abbr>

Represents an abbreviation/acronym; the title attribute can provide the full form.

<code>

Displays a short piece of programming code, usually in a monospace font.

<pre>

Preserves spaces and line breaks as written.

23. Additional Important Attributes

readonly

Allows users to view a field but not edit its value.

disabled

Disables an element so the user cannot interact with it.

maxlength

Limits the maximum number of characters.

minlength

Specifies the minimum number of characters.

value

Provides a value for a form control.

checked

Pre-selects a checkbox or radio button.

selected

Pre-selects an option in a <select>.

multiple

Allows multiple file selections or multiple options where supported.

autocomplete

Controls browser autocomplete behavior.

autofocus

Automatically focuses a form control when the page loads.

24. HTML <meta> Tag

What is <meta>?

The <meta> tag provides metadata about the HTML document. Metadata is not displayed as normal page content and is placed inside <head>. It is an empty element.

charset

Specifies character encoding. Example: <meta charset="UTF-8">.

viewport

Helps responsive layouts: <meta name="viewport" content="width=device-width, initial-scale=1.0">.

description

Provides a short page description that may be used by search engines.

keywords

Specifies keywords associated with a page; modern search engines generally do not use this tag as a major ranking signal.

author

Specifies the author metadata.

refresh

Can refresh or redirect after a specified time. Example: <meta http-equiv="refresh" content="5;url=https://www.google.com">.

Complete Example

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Student Portfolio Website">
  <meta name="keywords" content="HTML, CSS, Portfolio">
  <meta name="author" content="Student">
  <title>Student Portfolio</title>
</head>
<body>
  <h1>Welcome to My Portfolio</h1>
  <p>This is my first HTML webpage.</p>
</body>
</html>
```